11. Sequences and Series

11.6. Ratio and Root Tests for Series

11.6 Learning Objectives
- Know when and how to use the Ratio Test
- Know when and how to use the Root Test
- Understand why the ratio test is inconclusive when the ratio is 1
- Understand why the root test is inconclusive when the root is 1

For the two comparison tests, we checked for convergence by looking at the rates of growth for both series. For the Ratio Test, we are going to directly check the growth rate of the terms in the sequence to determine if the series (adding up the terms of the sequence) converges. We will compare $|a_{n+1}|$ to $|a_n|$, for ratio test we will do this by taking their ratio...

11.6.1. Ratio Test

Theorem 11.6.1 The Ratio Test

For a series $\sum_{n=1}^{\infty} a_n$

(i) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** (and therefore convergent).

(ii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **divergent**.

(iii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L = 1$, then the Ratio Test is **inconclusive**. The series $\sum_{n=1}^{\infty} a_n$ may absolutely converge, conditionally converge, or diverge. We cannot determine this with ratio test.

Note

Recall: For any geometric series,

$$a + ar + ar^2 + \ldots = \sum_{n=1}^{\infty} ar^{n-1}$$

the variable r is called the ratio. In fact, for the series, $r = \left|\frac{a_{n+1}}{a_n}\right|$, and if $|r| < 1$ the geometric series converges, otherwise it diverges.

For $r = 1$, the conclusion that the geometric series diverges does not come from ratio test. It is coming from the divergence theorem (another test).

Example 11.6.1

Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{5^n}$ is absolutely convergent, conditionally convergent, or divergent.

Note

- This is an alternating series, but the absolute value series, $\sum_{n=1}^{\infty} \frac{n^2}{5^n}$ is neither a p-series, geometric series, or something we want to integrate. It is also no a series we can easily compare to p-series or geometric series, so we try Ratio Test. Powers, exponentials, and (we'll later see) factorials can simplify a bit when we use Ratio Test.

Solution

By Ratio Test,

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(-1)^{n+1}(n+1)^2}{5^{n+1}}\right| \cdot \left|\frac{5^n}{(-1)^n n^2}\right| \\
&= \lim_{n\to\infty}\left|\frac{-(n+1)^2}{5n^2}\right| \\
&= \lim_{n\to\infty}\frac{1}{5}\frac{(n+1)^2}{n^2} \\
&= \lim_{n\to\infty}\frac{1}{5}\frac{n^2+2n+1}{n^2} \\
&= \lim_{n\to\infty}\frac{1}{5}\left(1+\frac{2}{n}+\frac{1}{n^2}\right) \\
&= \frac{1}{5} < 1 \checkmark
\end{aligned}$$

Therefore, by the Ratio Test, $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{5^n}$ converges absolutely.

Note: The "By Ratio Test" at the start isn't necessary since we say it again in the conclusion, it just feels weird jumping straight into the ratio without motivation.

Example 11.6.2

Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{(n^2+1)3^n}{n!}$ converges or diverges.

Note

We see exponentials and exponents and factorials all over the place! No other tests are equipped to deal with all of these at once $\Rightarrow$ **Try Ratio Test!**

Solution

By Ratio Test,

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{((n+1)^2+1)3^{n+1}}{(n+1)!}\cdot\frac{n!}{(n^2+1)3^n}\right| \\
&= \lim_{n\to\infty}\frac{((n+1)^2+1)\cancel{3^n}3^1\cancel{n!}}{(n+1)\cancel{n!}(n^2+1)\cancel{3^n}} \\
&= \lim_{n\to\infty}\frac{3(n+1)^2+3}{(n+1)(n^2+1)} \\
&= \lim_{n\to\infty}\frac{3n^2+6n+6}{n^3+n^2+n+1} \\
&\overset{\text{L'H}}{=} \lim_{n\to\infty}\frac{6n+6}{3n^2+2n+1} \\
&\overset{\text{L'H}}{=} \lim_{n\to\infty}\frac{6}{6n+2} \\
&= 0 < 1\checkmark
\end{aligned}$$

Therefore, $\displaystyle\sum_{n=1}^{\infty}\frac{(n^2+1)3^n}{n!}$ converges by the Ratio Test.

Remark

Notice that we need to be exponential experts and factorial fans! Brush up on your rules for both of these if needed!

Example 11.6.3

Try applying Ratio Test to determine the convergence of $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}$ and $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}$. What do you observe? Compare the results from Ratio test to what we know about these series as p-series.

Example 11.6.4

Use the ratio test to determine whether $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{n^n}$ is absolutely convergent, conditionally convergent, or divergent.

Note

We once again see exponentials and exponents and factorials all over the place! No other tests are equipped to deal with all of these at once ⇒ **Try Ratio Test!**

Solution

By Ratio Test,

$$\begin{aligned}\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(n+1)!}{(n+1)^{n+1}}\cdot\frac{n^n}{n!}\right| \\ &= \lim_{n\to\infty}\frac{(n+1)n^n}{(n+1)^{n+1}} \\ &= \lim_{n\to\infty}\frac{n^n}{(n+1)^n} \\ &= \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n\end{aligned}$$

This limit is a little tricky, but it has indeterminate form 1^∞ so we need to use our exponential and log trick. Consider

$$\begin{aligned}\lim_{x\to\infty}\ln\left(\left(\frac{x}{x+1}\right)^x\right) &= \lim_{x\to\infty} x\ln\left(\frac{x}{x+1}\right) \\ &= \lim_{x\to\infty}\frac{\ln\left(\frac{x}{x+1}\right)}{\frac{1}{x}} \quad \text{has IFL: } \frac{0}{0} \\ &\overset{\text{L'H}}{=} \lim_{x\to\infty}\frac{\frac{x+1}{x}\frac{x+1-x}{(x+1)^2}}{-\frac{1}{x^2}} \\ &= \lim_{x\to\infty} -\frac{x}{x+1} \\ &= -1\end{aligned}$$

So

$$\lim_{x\to\infty}\left(\frac{x}{x+1}\right)^x = e^{\lim_{x\to\infty}\ln\left(\left(\frac{x}{x+1}\right)^x\right)} = e^{-1}$$

So then

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n = \frac{1}{e} < 1.$$

Therefore, $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{n^n}$ is absolutely convergent by the Ratio Test.

11.6a Section Summary:

- We went over ratio test and when to use it.